Somnium: A Recurrent World Model that Learns to Dream Before It Wakes

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Abstract

We train a small non-Transformer world model, SOMNIUM, under a developmental curriculum: *chaos* → *dream* → *wake*. A single recurrent substrate carries two dynamics—fast streaming expansion (*yang*) and energy-relaxation contraction (*yin*)—softly mixed by a gate; prediction happens in latent space (JEPA-style next-latent), and training begins with a *dream phase* of self-generated rollouts before any real environment interaction. Dream-first pretraining reliably cuts the real-interaction steps needed to reach a fixed accuracy by ∼60%: replicated across five independent runs at 16K parameters (ratios 0.39–0.42 against a pre-registered threshold of 0.6). A scale ladder, a dose-scaling control, and autonomous growth runs converge on the same conclusion: the advantage is *capacity-bound*. This paper then documents what happened when we tried to break that boundary and, in the process, broke our own instruments instead. A five-hypothesis exclusion chain traced the wide-model pathology ($d_h \geq 256$) to optimization geometry—not limited expressivity—and a single output LayerNorm cures it (plateau MSE 1.03 → 0.17 at d_h=1024; the width cliff flattens to 0.115–0.175 across 64–1024). Two instrument defects surfaced and were repaired under pre-registration: JEPA mean-squared error is a *moving yardstick* (encoder target scale drifts 11× across configurations; we switch to scale-invariant decode fidelity), and freshness-only replay causes systematic recency drift (fixed by mixed replay: 0.415 → 0.011 at 9000 cycles). Re-running the central question on the repaired instrument *re-establishes the capacity-bound conclusion in cleaner form*: the dream advantage survives only at the width where wake itself is capacity-limited (d_h=64; ratios 0.58/0.69), vanishes at 256–1024 (0.74–0.98 or unreached), and the transferable component is *replay*—once every schedule has replay access, the full dream battery adds nothing. All hypotheses, thresholds, corrections, and outcomes—including the retraction of our own earlier "expressivity ceiling" claim—are logged in a public run registry. A theory that can be killed is a theory that can evolve.

Keywords: world models; pretraining; self-supervised learning; continual learning; sleep-inspired computation; pre-registration; negative results

1 Introduction

Should a model meet the world awake or asleep? Mammalian development suggests an ordering: spontaneous, self-generated activity (retinal waves, hippocampal replay) precedes and shapes later sensorimotor learning [1, 2]. Sleep research offers candidate mechanisms—synaptic homeostasis [3], reverse learning [4], consolidation via replay [5]. In machine learning, world models trained by latent imagination [6] and self-supervised objectives in representation space [7, 8] echo pieces of this story, but the developmental question—*does a model that dreams before it wakes learn more efficiently from its later waking data?*—is rarely asked directly.

We built SOMNIUM to ask it under a discipline borrowed from clinical research: **pre-registered ablations with a public outcomes registry**. Every claim in this paper was registered with

a threshold before the run that tested it; every negative result is retained and reported. The contributions are:

1. **Architecture** (§2). A non-Transformer world model: one recurrent substrate with dual dynamics (streaming expansion / energy relaxation), gate-mixed; JEPA-style next-latent training with $O(1)$ state; a function-preserving width-growth operator; parameter count $3d_h^2 + O(d_h)$.
2. **A replicated, capacity-bound positive result** (§3). Dream-first pretraining reaches a wake-only model's final accuracy with ∼40% of the real interactions, replicated 5× at 16K parameters across two compute platforms. A scale ladder (16K/210K/3.2M), a dose-scaling control (4×/16× dream budget), and autonomous growth runs independently agree: the advantage is real but *capacity-bound*, consistent with its mechanism being energy smoothing—an initialization effect that large models do not need.
3. **A registry of pre-registered negative results** (§4). VQ situation codebooks hurt training; learned gating gives no accuracy gain (but its polarity is an emergent developmental marker: a one-way ratchet from corrective to feed-forward regimes); dream-statistics feedback does not improve calibration.
4. **An autonomous overnight protocol** (§3.4) in which the model trains itself through self-probes, expands its own width by dreaming, and—in the second night—detects its own numerical divergence and rolls back to its last healthy self.

We emphasize what this paper is *not*: SOMNIUM is a 16K–3.2M-parameter model on procedural gridworlds. Nothing here claims general-purpose intelligence. The point is methodological: with pre-registration and honest registries, a two-day single-GPU project can produce claims that replicate, boundaries that are measured rather than assumed, and failures that accumulate into design knowledge.

2 Architecture

2.1 One substrate, two dynamics

SOMNIUM has no attention and no token sequence. Its core is a single latent state $h_t \in \mathbb{R}^{d_h}$ updated once per environment step by soft-mixing two dynamics that *share all parameters* (a deliberate contrast with dual-network designs; cf. the finding that a single tiny recursive network outperforms a biologically-inspired hierarchy [9]):

$$h_{t+1} = h_t + \alpha_t \, \Delta_t^{\text{yang}} + (1 - \alpha_t) \, \Delta_t^{\text{yin}} \tag{1}$$

$$\Delta_t^{\text{yang}} = \text{dt} \cdot \tanh\big(W_c[h_t; e(o_t); a_t]\big) \tag{2}$$

$$\Delta_t^{\text{yin}} = -\text{lr} \cdot \nabla_h E(h)\big|_{h=h_t}, \qquad \alpha_t = \sigma\big(g([h_t; x_t])\big) \tag{3}$$

where $e(\cdot)$ is an encoder shared with the training target, W_c a linear cell, E a small MLP energy head, and g a linear gate initialized at $\sigma(0) = 0.5$ (born half-and-half). Equation (1) is the model's entire per-step computation: the recurrent state is $O(1)$, "thinking longer" costs compute but never parameters, and the parameter count is $3d_h^2+52d_h+296$ for our 8-dimensional observations and 4 actions—e.g. 16K at d_h=64, 3.2M at d_h=1024.

2.2 Training objective

Training is JEPA-style prediction in latent space, never next-token and never pixel reconstruction:

$$\mathcal{L} = \big\|h_{t+1} - \text{sg}\big(e(o_{t+1})\big)\big\|^2 + \lambda_r\big(\hat{r}_{t+1} - r_{t+1}\big)^2 + \lambda_v \, \mathcal{N}\big(\hat{r}_{t+1}, \hat{\sigma}^2_{t+1}; \, r_{t+1}\big) + \lambda_d\big(\hat{o}_{t+1} - o_{t+1}\big)^2, \tag{4}$$

with sg(·) a stop-gradient on the target (v0; no EMA teacher yet), $\hat{\sigma}$ a heteroscedastic uncertainty head used only for probes and abstention experiments, and $\hat{o}$ a low-weight observation decode used only to close the loop for planning probes. Each training sequence begins from a noise sample (the "chaos" stage): $h_0 \sim \mathcal{N}(0, I)$.

2.3 The dream phase

Before the first real interaction (*dream-first*) the model runs self-generated rollouts with four independently togglable components: **imagination** (latent rollouts from the noise prior; the effective term turns out to be energy smoothing, §5), **consolidation** (replay distillation from the episode buffer), **reverse learning** (noise-augmented replay, after [4, 10]), and **rehearsal** (oversampling rare transitions). Three schedules are compared throughout: *wake-only*, *dream-first* (dream pretraining, then wake-only), and *alternate* (a dream phase after each wake cycle). A fairness note applies to all comparisons: dream schedules receive extra offline gradient steps; the metric is always *real environment interactions*.

2.4 Function-preserving growth

To let the model widen itself, we use a Net2Net-style operator [11]: new latent dimensions copy a random source dimension in all *producer* weights (encoder output, cell output) while all *consumer* weights receive zero columns. At zero noise the update map is bit-exact before and after growth (verified in tests); with small noise on new rows only, perturbation is $O(\varepsilon)$ and new dimensions wake up through gradients. Growth multiplies the dream budget by 1.5 ("the dream deepens as the dreamer grows"); the optimizer is rebuilt at each growth event.

3 Experiments

3.1 Protocol

Environments are procedural gridworlds (5–12 cells; 8-dim observations: position, goal and hazard offsets, clock; slippery actions; rare hazard teleports) and DFS mazes. All comparisons hold the wake data stream identical across schedules. The primary metric is the *steps-to-target ratio*: real interaction steps for a schedule to first reach the wake-only model's final accuracy, divided by the wake-only total. Thresholds were committed before each run: ratio ≤ 0.6 (i.e., a $\geq$40% saving) counts as positive. Code, registry, and per-run records: `https://github.com/wilberforce1900/somnium`.

3.2 Dream-first cuts real interactions by 60% (at 16K)

In the base configuration (d_h=64, ∼5.3K real steps total), dream-first reaches wake-only's final latent MSE using 39%, 39%, and 42% of the real interactions (three seeds; final MSE 0.235 vs. 0.905 at equal steps). A cross-platform replication on an L4 GPU reproduced 0.41/0.39 with the identical configuration—a bit-exact determinism check. The *alternate* schedule shows the same direction (0.77) without passing threshold: the benefit concentrates in *pre-training*, not in interleaving. Component decomposition (six variants, pre-registered) attributes the sequential-learning benefit to replay components (consolidation +6.1pp, reverse learning +6.0pp, rehearsal +5.3pp on old-task improvement) while imagination contributes nothing there—its contribution is confined to the birth-time sample efficiency above, a division of labor we summarize as *imagination is for being born; replay is for keeping what waking taught.*

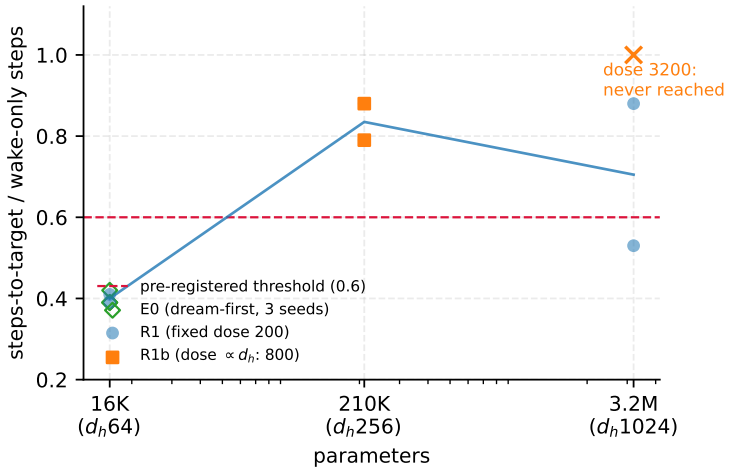


Figure 1: **Dream advantage is capacity-bound.** Steps-to-target ratio (lower is better; pre-registered threshold 0.6 dashed). At 16K parameters dream-first reaches wake-only's final accuracy with ∼40% of the interactions (five independent runs across two platforms). At 210K/3.2M the advantage fades; scaling the dream budget with width (R1b) does not restore it, and at the largest scale the highest dose never reaches the target.

3.3 The advantage is capacity-bound

Figure 1 shows the scale ladder ($d_h \in \{64, 256, 1024\}$, two seeds each): ratios 0.41/0.39, 0.88/0.79, 0.88/0.53. Two confounds were closed by pre-registered controls:

Dose. Because the dream budget was fixed at 200 phases while width grew, we re-ran with dose $\propto d_h$ (200/800/3200). Outcome: unchanged at 210K (0.88/0.79, bit-identical trajectories' summary values), and at 3.2M the 16× dose *never reaches* the target (slightly worse than fixed dose). Under-dosing is not the explanation.

Task saturation. Wake-only final accuracy is flat in width (MSE 0.89/0.98/1.00): the environment family saturates near a few hundred dimensions, compressing any sample-efficiency advantage.

3.4 Dream-grown models hit the same wall

In autonomous overnight runs the model chose its own environment difficulty from four self-probes (normalized residual $\text{res}_{\text{norm}} = \|h - e(o')\|/\sqrt{d_h}$, uncertainty $\hat{\sigma}$, coverage, gate polarity α) and grew its width after two consecutive improving rounds. Night 1 (871 rounds, no safeguards) grew healthily from 64 to 486 (res_{norm} band 0.29–0.60), then the step to 1024 exploded the residual 6.8×; 667 subsequent rounds of self-repair halved it without recovery, with the gate pinned at the corrective pole ($\alpha \approx 0$) throughout. Night 2 (guard rails: growth only when $\text{res}_{\text{norm}} < 0.6$, an anti-starvation rule for environment choice, gradient clipping, and rollback-to-last-healthy-self on divergence) grew to 729 while healthy, explored the environment grid to its ceiling, and—after an unclipped divergence late in the night—*detected the failure via its own* $\hat{\sigma} \to \infty$ *probe and restored itself from its round-800 checkpoint.* Night 2 was then extended and ran to 4,932 rounds total (night 1's 5.7×): at d_h=1024 the model trained *stably for 3,000+ rounds* inside the res_{norm} band 0.57–0.81, slowly converging to 0.73 with zero rollbacks—the clipping and guards fully cured night 1's failure mode at the same scale. Figure 3 shows both trajectories.

The wall moves: from expressivity to optimization geometry. Three probe experiments first triangulated where the wall sits: the flat scale ladder (§3.3); a 90-cycle plateau test at d_h=1024 (3× budget) confirming the flatness is not under-training; and deepening the BPTT

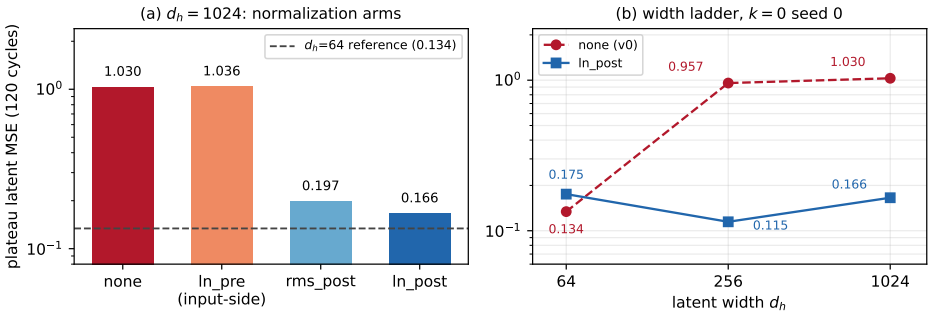


Figure 2: **The width cliff was an optimization defect, not an expressivity ceiling.** (a) Plateau MSE at d_h=1024 (log scale; 120-cycle plateau-aligned protocol, seed 0): output normalization cures the 1.03 floor (ln_post 0.166, rms_post 0.197; "cured" threshold ≤ 0.30 registered pre-run), input-side normalization (ln_pre 1.036) does not—the pathology is in the state write, not the cell read. Dashed: the d_h=64 reference 0.134. (b) Width ladder: the cliff (0.134/0.957/1.030) flattens under output LayerNorm (0.175/0.115/0.166).

horizon ($T \in \{8, 32, 64\}$) *degrading* training monotonically (0.67/1.37/2.80). A pre-registered exclusion chain then eliminated the remaining suspects one by one—training budget, BPTT length, encoder rank, learning rate, and cell expressivity (a three-cell comparison: GRU-style gating 1.081, two-layer 1.017, vs. linear 1.030; best improvement 1.2%)—leaving optimization geometry itself. The decisive experiment was the cheapest: normalizing the latent state at the *output* of the mixed update (width-free, no new state; LayerNorm [12] or RMSNorm [13]). Figure 2 shows the outcome at d_h=1024: output LayerNorm reaches 0.166 and RMSNorm 0.197 against the pathological floor 1.03 (threshold for "cured": ≤ 0.30, registered before the run; the d_h=64 reference is 0.134), while *input-side* normalization (1.036) does nothing—the pathology lives in the *write side*, the accumulation of state, not in what the cell reads. The mechanism panel explains why: without normalization the gate *collapses* early ($\alpha \to 0.016$: the yang pathway shuts off; the yin step is two orders of magnitude smaller) and the state stays frozen at its birth scale—the "floor" was simply the variance of the spawn distribution, never the target's. Output normalization releases the trap: the gate recovers ($\alpha \to 0.999$), the state scale becomes a learned quantity that tracks the target manifold, and the width cliff flattens (0.175/0.115/0.166 at $d_h = 64/256/1024$, vs. 0.134/0.957/1.030 without). One caveat is kept on the books: RMSNorm's cure is seed-sensitive (0.197 vs. 0.372; a second seed falls back into the collapse trap and escapes late), so LayerNorm is the robust fix. *We therefore retract the earlier claim that a single linear cell imposes a hard ceiling near a few hundred dimensions*: the $d_h \leq 256$ applicability domain was an artifact of a fixable optimization defect. The five negative probes stand as facts; the inference drawn from them did not.

3.5 A moving yardstick: converged protocols and scale-invariant metrics

The healing forced an audit of the measuring instruments themselves, with two findings that change how every number in this paper should be read. First, **JEPA mean-squared error is a moving yardstick**: the encoder defines its own target space, and that space's scale drifts during training (per-dimension target variance differs by 11× across converged configurations). Cross-configuration comparisons of raw latent MSE are therefore meaningless; we re-anchor on *decode fidelity*—the R^2 of decoding the next observation from the latent state—against a copy-the-current-frame baseline. Second, **short budgets measure "not started"**: a budget bridge at d_h=256 on a rich world (partial observations, key–door dynamics, four rooms) shows JEPA R^2 at −48% after 120 cycles (worse than predicting the dataset mean), +64% after 1,000, and decode R^2 climbing 51% → 87% → 93% (saturating near 4,000 cycles). Every "plateau" reported

by the 120-cycle calibration protocol was a snapshot taken before learning began; all v1.0/v1.1 absolute MSE values are retained in the registry as historical record under this warning.

On the repaired instruments the width question re-opens cleanly. Long runs (9,000 cycles, ∼12M interactions) on the richest world show wide margins: the d_h=1024 model descends to 0.017 latent MSE where its 120-cycle snapshot read 0.138, and decode fidelity reaches 93%—seven to ten times better than a frame-copy baseline (30%), i.e. genuine next-state prediction, not persistence. The width ladder at convergence reads 92.6% (d_h=256) ≈ 93.2% (1024) > 90.2% (2048): width parity across 256–1024, with 2048 a slow starter (120-cycle MSE 0.96, i.e. the same pathology signature) that converges normally—not a trap. A second instrument defect surfaced in these long runs: with freshness-only replay (sampling only the newest episodes), the d_h=1024 model systematically drifts toward recent data (0.118 at mid-training → 0.415 at the end, reproduced across seeds). A pre-registered fix—mixed replay, half fresh windows and half uniform over the whole buffer—cures it (0.011 final; gate value 0.20 below the 0.30 failure bound, 39× below the drift itself). Mixed replay is the standard protocol for everything that follows.

3.6 The central question, re-run on the repaired instrument

With a healthy architecture (output LayerNorm), a converged protocol (1,200 cycles), and mixed replay for all schedules, we re-registered the original A3 comparison: three schedules × three widths × two seeds, threshold unchanged (steps-to-target ≤ 0.6× wake-only's total, anchored *within* the matrix at each width's wake-only converged accuracy):

d_h	seed	wake final	dream-first	alternate	verdict
64	0	0.051	**0.58** ✓	**0.56** ✓	borderline pass
64	1	0.043	0.69 ×	0.62 ×	borderline fail
256	0	0.018	unreached	unreached	fail
256	1	0.019	0.74 ×	unreached	fail
1024	0	0.018	0.77 ×	unreached	fail
1024	1	0.016	0.98 ×	unreached	fail

The capacity-bound conclusion returns, sharper than before. The wake-only ladder itself carries the signature: d_h=64 caps at 0.043–0.051 while 256/1024 converge to 0.016–0.019—and the dream advantage (now marginal: 0.56–0.69 where it was 0.39–0.42 on the old instrument) appears *exactly* at the width where wake itself is capacity-limited, vanishing where capacity suffices. One comparison reframes the mechanism: the *alternate* schedule—the full four-component dream battery—never beats wake-only *with mixed replay*. Replay was the transferable component all along; once every schedule has access to it, imagination adds only the birth-time margin at the smallest width. "Imagination is for being born; replay is for keeping what waking taught" survives its sharpest test.

4 Negative results registry

All thresholds were pre-registered; all results retained.

A1 — 64-entry VQ situation codebook ("hexagram" discrete planning layer). Motivated by treating situations as a discrete transition graph over which a slow planner searches (value iteration over 64 codes, Xiantian binary encoding). Outcome at two seeds: the *baseline without a codebook* reaches 0.406 success on long-horizon navigation while every codebook configuration scores ≤ 0.094; ablations show the VQ commitment loss *damages the continuous model during training*. Two planner implementations (direct code scoring; potential-shaped search [14]) failed a pre-launch check against an equal-compute control. Demoted to a read-only interpretability layer.

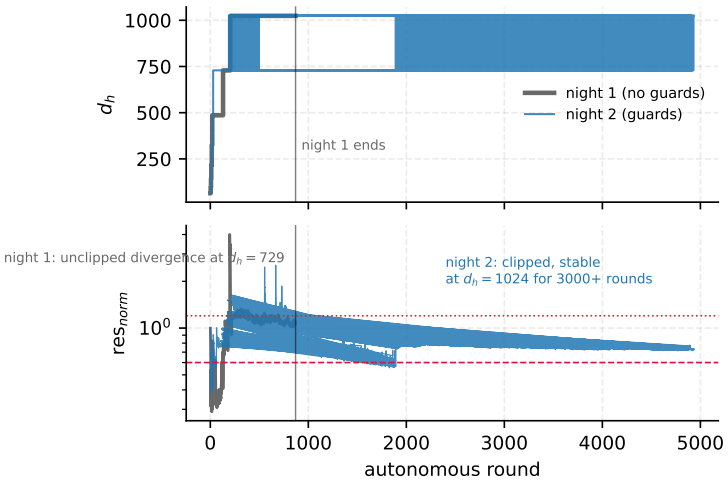


Figure 3: **Two nights of autonomous self-training.** Top: width d_h over rounds (growth events). Bottom: normalized residual (log scale). Night 1: healthy growth to 486, destabilization at 1024, 667 rounds of unrecovered self-repair. Night 2 (guards + clipping + rollback): growth to 729 while healthy; the divergence near round 870 is rolled back to the last healthy self (round 800) and training continues below the band.

A2 — learned dual-mode gating. Learned gate vs. fixed single-mode at equal parameters: +0.9pp / −0.8pp on two tasks against a +2pp threshold. The gate *learns* (it settles at $\alpha \approx 0.3$) but converts to no accuracy. The interesting residue is what the gate encodes: polarity is *co-adapted* (forcing a wake-trained model to the yang pole at inference inflates prediction error 4.4×), and it behaves as a **one-way developmental ratchet**—both schedules slide to the corrective (yin) pole early in training ($\alpha \to 0.17$); dream-pretrained models then ratchet to the feed-forward pole ($\alpha \to 0.96$) while wake-only models stay corrective; injecting a new task does *not* reopen the gate. The gate is a stage marker, not an online "do-I-know-this" meter: a falsification we ran before believing it.

A5 — reading dreams to guide sampling. Feeding dream-log statistics back (coverage-blind-spot bias of imagination; inverse-coverage-weighted replay) against matched controls: first implementation was mechanically inert (the imagination objective is content-insensitive); the second verifiably changed sampling but produced *zero* calibration gain. Root cause: the uncertainty head is calibrated out of the box (95% interval coverage 95–97%), leaving no headroom to improve; per-window error discrimination was unstable across seeds (−0.12 vs. +0.72). Demoted to an observation layer.

5 Discussion

Mechanism. Ablations localize the dream-first benefit to the energy-smoothing term of imagination (the anti-collapse variance floor was provably never active): pretraining flattens the energy landscape that the yin dynamics descend, i.e., it shapes the geometry the wake phase will optimize in. This is an initialization effect, which is exactly why it is capacity-bound—large models buy the same geometry with data. The instrument-repaired replication (§3.6) sharpens this twice over: the margin survives only at the width whose wake phase is itself capacity-limited, and the interleaved dream battery never beats wake training that merely has replay access—the transferable component of dreaming here is *replay*, and imagination's residue is a birth-time margin for small models. The alignment with sleep theory is comfortable but we state it flatly: what we implement is energy smoothing plus replay [3, 4, 10]; no claim about consciousness is made or implied, and we regard "AGI as the awakening of silicon life" as poetry, not prediction.

Boundaries. Toy environments; two seeds at scale; no external architecture baseline (the natural next control is a parameter-matched small transformer); compute asymmetry between dream and wake schedules is documented rather than eliminated; the registry contains two infrastructure failures (a shell incompatibility that produced a false completion, and an unbounded-checkpoint disk exhaustion) that consumed real hours and are reported because they will consume someone else's. One further boundary is self-inflicted and worth stating plainly: every absolute MSE value in §3 was produced by the pre-repair instrument (under-trained budgets, drifting yardstick, freshness-only replay); we retain them as registered history and read only their relative structure.

The ceiling as a finding—and its correction. We initially read the capacity wall as a limitation of the environments, designed richer ones to escape it, then—when the wall persisted—read it as an expressivity ceiling of the single linear cell. Both readings were wrong, and the correction is itself part of the record: a five-hypothesis exclusion chain (all negative) followed by a four-arm normalization study (two cures, one inert arm localizing the defect to the write side) moved the wall from *expressivity* to *optimization geometry*, retracted the $d_h \leq 256$ applicability claim, and restored wide models to small-model accuracy with one width-free layer. We keep the section heading's original claim in the registry rather than editing it away, because the sequence is the point: pre-registered claims are cheap to state, cheap to falsify, and—as here—cheap to correct in public. What survives the correction is sharper than what preceded it: the dream advantage tracks the model–world capacity match with a precision the old instrument could not see.

What would change our mind. The capacity-bound claim remains falsifiable in both directions. A world whose decode-fidelity ladder keeps rising with width at converged protocol (our richest current world shows 256–1024 parity, so today's worlds do not feed wider bodies) would re-open the question of whether dreaming returns when there is structure worth dreaming about. The non-stationary axis—the sleeping brain's home turf—is now scoped by a clock-calibration pilot: a single hidden regime switch between two slip dynamics measures adaptation time (τ_{50} = 120–312 cycles), but the model *pre-generalizes* across such mild regimes and shows zero retroactive interference, so prediction-layer slip switches cannot test the hypothesis; the registered next step is strategy-layer regime differences where the optimal policy itself flips. That experiment—not a bigger model on the same gridworlds—is where our registry points next.

6 Related work

World models with latent imagination [6] train policies on self-generated rollouts; Somnium differs in training the *world model itself* before its first real step, under a developmental schedule. JEPA [7, 8] and latent-reasoning lines [15, 16] predict in representation space; we adopt the objective wholesale. Our growth operator follows Net2Net function-preserving widening [11]; progressive growing has a long history [17]. Sleep-inspired ML ranges from wake-sleep algorithms [18] to replay for continual learning [10]; Somnium imports the vocabulary (homeostasis [3], reverse learning [4]) as engineering hypotheses subject to the same ablations as everything else. Finally, the pre-registration discipline follows clinical-trial practice; to our knowledge its combination with a public per-run registry is uncommon in small-scale ML research, and we found it changes what one is willing to claim.

7 Conclusion

A two-day, single-GPU project; five replications of a 60% interaction saving; three independent measurements of where that saving ends; three pre-registered negatives kept on the books; a model that, left alone overnight, grew itself, hit its own wall, and learned to restore its last

healthy self—and then, on a repaired instrument, the same question asked again and answered more precisely: the saving lives only where capacity is binding, and replay is the load-bearing component. Along the way the paper retracts one of its own registered conclusions (an expressivity ceiling that turned out to be a one-layer fix) and logs the retraction with the same discipline as the original claim. We offer SOMNIUM less as an architecture than as a working argument that the unit of progress in small-scale ML research can be the *registered, falsifiable, retained claim*—including the ones that die, and the ones we kill ourselves.

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